

The Earth-Our World

Our Earth & Other Planets of Solar System

- Ravivar, Mangalvar, Shukravar, or Shanivar are some of the names of the days of the week.
- It is interesting to know that these names have been derived from heavenly bodies.
- The Sun (ravi), Mars (mangal), Venus (shukra), and Saturn (shani) are all heavenly bodies that can be seen from the earth.
- When man first landed on moon (July 19, 1969) the earth was seen as a shining spherical ball, the way we see other heavenly bodies.



Earth as seen from Moon.

The Shape of Our Earth

- Ordinarily, the earth appears flat to us. That is so because we can, at any point of time see only a very small part of it.
- Remember the Portuguese navigator Magellan, who sailed around the world and proved that the earth is round.
- It should, however, be remembered that our earth is not exactly like a sphere. More like an orange, it is flat at the poles.



The shape of our Earth is not round but it is like an Orange: Flat at Poles.

Continents of Earth

- There are seven large masses of land called Continents. Asia is the largest of them all.



Seven Continents of Earth

- India is a part of the continent of Asia. All continents are divided into smaller units called countries.
- Antarctica is one continent where there is no habitation except that of visiting scientists.
- Asia and Europe are separated by the Ural mountains. Australia is the only continent that is made up entirely of islands.
- North America and South America are linked by Central America, which forms the Southern part of North America.
- The continent of Africa was at one time known as the Dark Continent.

Water Bodies on Earth

- There are large masses of water on earth called oceans. One of them is named after a country which is Indian Ocean.



Oceans and Continent of Earth

- The other oceans are the Pacific, Atlantic, Arctic, and Southern Ocean.
- Smaller bodies of water are called sea. Seawater is salty.



Difference Between Sea and Oceans

- There are lakes and rivers whose water is fit for drinking.
- The distribution of land and water on the surface of the earth can be easily seen on the globe.
- Similarly, we can have a good idea of the relative sizes of the various continents by looking at the globe.

Globe

- If we have to travel to the west coast of North America, we may cross the Pacific Ocean or the Atlantic Ocean. Which would be the shorter route? A look at the globe can provide an easy answer.



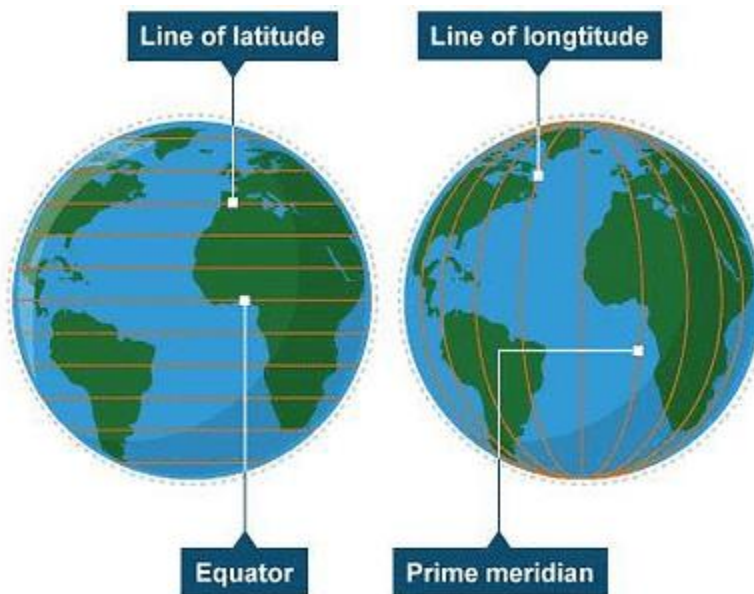
Globe

- The globe that you would see in your classroom is a sphere. It has an axis running through the two extreme ends–the North and the South poles.
- The globe can rotate around the axis. The axis is not exactly vertical. It is mounted on a horizontal base at a certain inclination.

Features of Earth as Studied through Globe

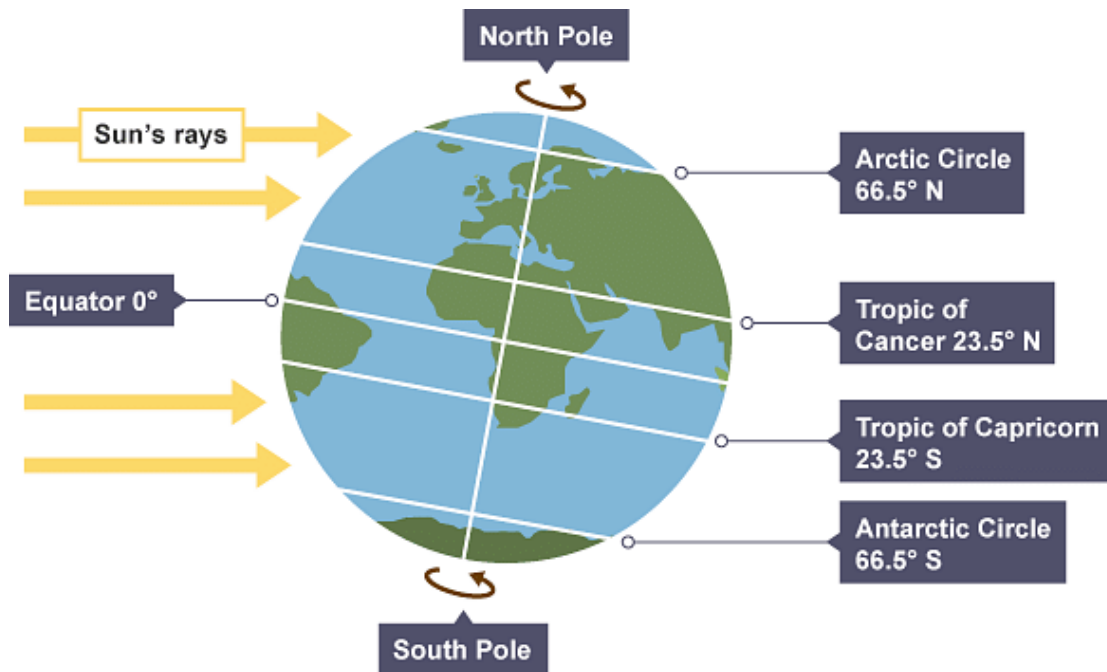
- A circle divides the globe into two hemispheres– Northern and Southern which is called equator. There are concentric circles on both sides of the equator.

Latitudes



Latitudes and Longitudes of Earth

- Their circumferences keep on diminishing with their distance from the equator. These circles are called Latitudes, North or South depending on which side of the equator they are placed.
- The Earth showing the Tropic of Capricorn at 23.5° South of the equator.
- The latitudes are marked from 0° to 90° N/S.
- The equator marks the latitude 0° while those at the poles mark 90° N or S. The latitude $23\frac{1}{2}^\circ$ N is called the Tropic of Cancer. The latitude $23\frac{1}{2}^\circ$ S is called the Tropic of Capricorn.
- The Tropic of Cancer runs through India. The two other latitudes with special names are the Arctic Circle ($66\frac{1}{2}^\circ$ N) and the Antarctic Circle ($66\frac{1}{2}^\circ$ S).

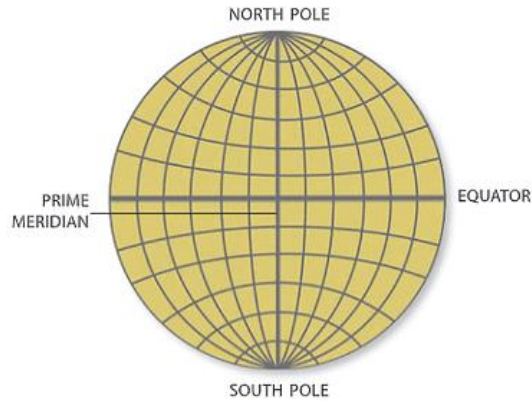


Longitudes

- The semicircular lines joining the poles are called Longitudes or Meridians. The Meridian passing through the Greenwich Observatory (near London in Britain) is taken as the reference point.
- It is the Prime Meridian.
- There are longitudes, marked 0° to 180° E and 0° to 180° W. The latitudes vary in circumference.
- The longitudes are all of equal length. All these are only imaginary lines drawn on the globe.
- If you visit Greenwich, you would find no such line physically marked on the earth's surface. The latitudes and longitudes make a network on the surface of the globe.

Grids

- The squares thus formed by intersecting Longitude and Latitude lines are called 'grid'.



Grid Lines of Globe

- Once we know about the latitude and the longitude nearest to any particular place, we can easily locate it on the globe.
- Pondicherry, in India, is situated near the latitude 12° N. It also lies near the longitude 80° E. With this information, it would be very easy to locate Pondicherry on the globe.
- The nearest latitude for Bhopal in Madhya Pradesh is the Tropic of Cancer. The nearest longitude is 78° E.
- The mountain top K2 (height 8611 m.) lies near the latitude 36° N. It lies within the grid formed by latitudes 34° – 36° N and the longitudes 76° – 78° E.
- These imaginary and artificial lines on the globe help us in the study of lands and people in many more ways.
- The location of a place in terms of meridian determines its local time. The latitudes influence the climates of the various regions on the earth.